

- **Revocable access:** Resource failure could be handled by granting access to other available resources.
- **Unlimited resource addition (scaling):** Being a distributed computing model, it allows easy growth of system capacity by adding more resources into an existing system.

Grid computing is sometimes labeled as the direct predecessor of cloud computing.

Table 2.1 Characteristics of cluster and grid computing

Characteristic	Cluster Computing	Grid Computing
Ownership	Single	Multiple
System management	Centralized	Decentralized
User management	Centralized	Decentralized
Resource Pool Creation	Yes	Yes
Resource node type	Homogeneous	Heterogeneous
Resource scheduling	Centralized	Decentralized
Network connection type*	LAN	LAN/MAN (Metropolitan area network)/WAN
System architecture	Single system image	Autonomous, independent nodes
Coupling	Tight	Loose
Fault tolerance Capability	Low	High
Scaling Architecture	Scalable	Scalable, in bigger capacity
Real-time Scaling*	No	No

*During the period of consideration

2.1.8 Hardware Virtualization: Weapon for Designing Dynamic

Computing System

and it is difficult to distinguish different characteristics of these three computing models. The following table presents a comparison between these three computing techniques.

Table 2.2 Comparison among cluster, grid and cloud computing

Cluster Computing	Grid Computing	Cloud Computing
A cluster is normally formed with computers of a single location, otherwise the system becomes complex.	Grid is inherently more distributed by its nature. The computers need not to be in the same geographical location.	It allows total distribution of resources like the grids. Hardware resources are maintained in multiple data centers spread across the globe.
Computation job takes place in one administrative domain owned by a single party.	Computation could occur over many administrative domains owned by multiple parties as connected together.	Computing resources of a cloud is usually owned by a single party. But multiple administrative domains can be combined together to perform the job.
In a cluster, all computing nodes should have similar hardware systems. That is, the system should be homogeneous in nature.	It can be heterogeneous in nature. The computers that are part of a grid can be made of different hardware architectures.	It can use heterogeneous collection of commodity hardware.
It features the centralized task management and scheduling system.	It features the distributed task management and decentralized scheduling.	It features the decentralized task management with more dynamic computing infrastructure.
Resources are generally pre-reserved for specific type of task.	Resources are generally pre-reserved for specific type of task.	Resources are not pre-reserved for specific task. Resource utilization is mainly demand-driven.
System is not dynamic in nature. Application mobility is not possible.	System is not dynamic in nature. Application mobility is not possible.	It is a dynamic system. Mobility of application is an inherent feature in this system.
One whole cluster behaves like a single system. As resources are managed by centralized resource manager, the individual computers can not be operated as separate computers.	Every node is autonomous that is, it has its own resource manager and behaves like an independent entity. So, each computer can be operated independently as distinct computer	There is no concept of directly accessing any particular physical computing nodes. Underlying computing infrastructure remains hidden from the users.

This chapter gives a broad idea about the foundation of cloud computing models. But it is very difficult to justify all the features of cloud computing in such a limited discussion. A detailed discussion is provided in subsequent chapters with proper explanations.

a generic one and not tied to any specific reference implementation or vendor's product. In addition to this, the model defines the actors, standard activities and functions associated with the cloud computing.

NIST defines cloud computing by describing five essential characteristics, three cloud service models and four cloud deployment models.

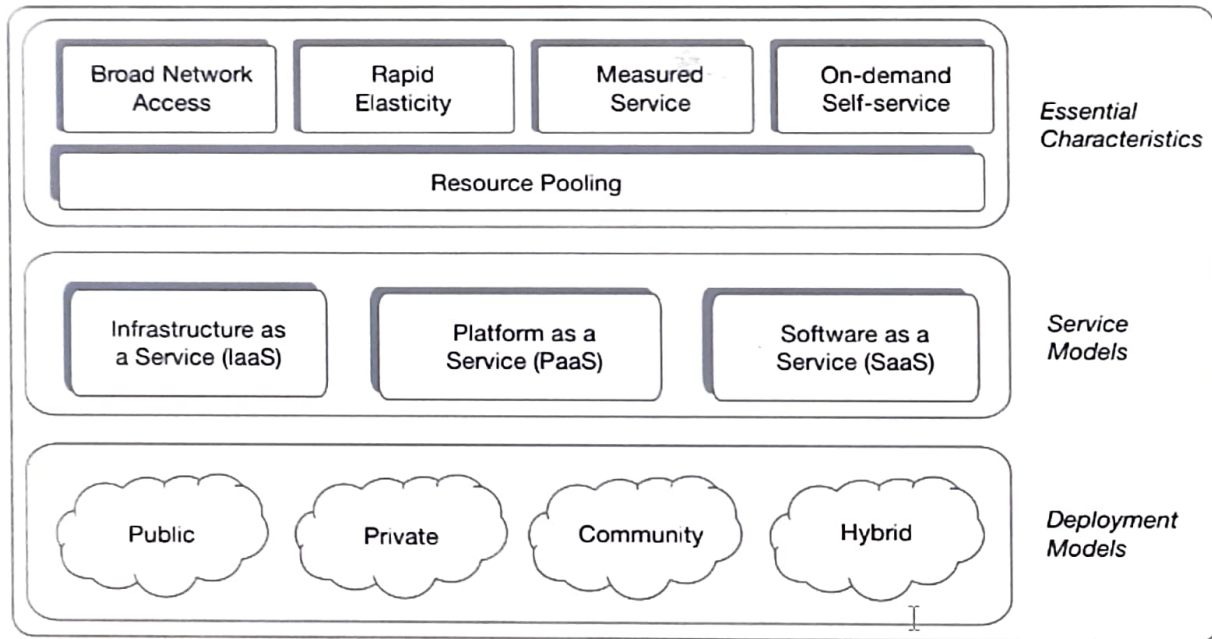


FIG 4.1: The NIST cloud computing model

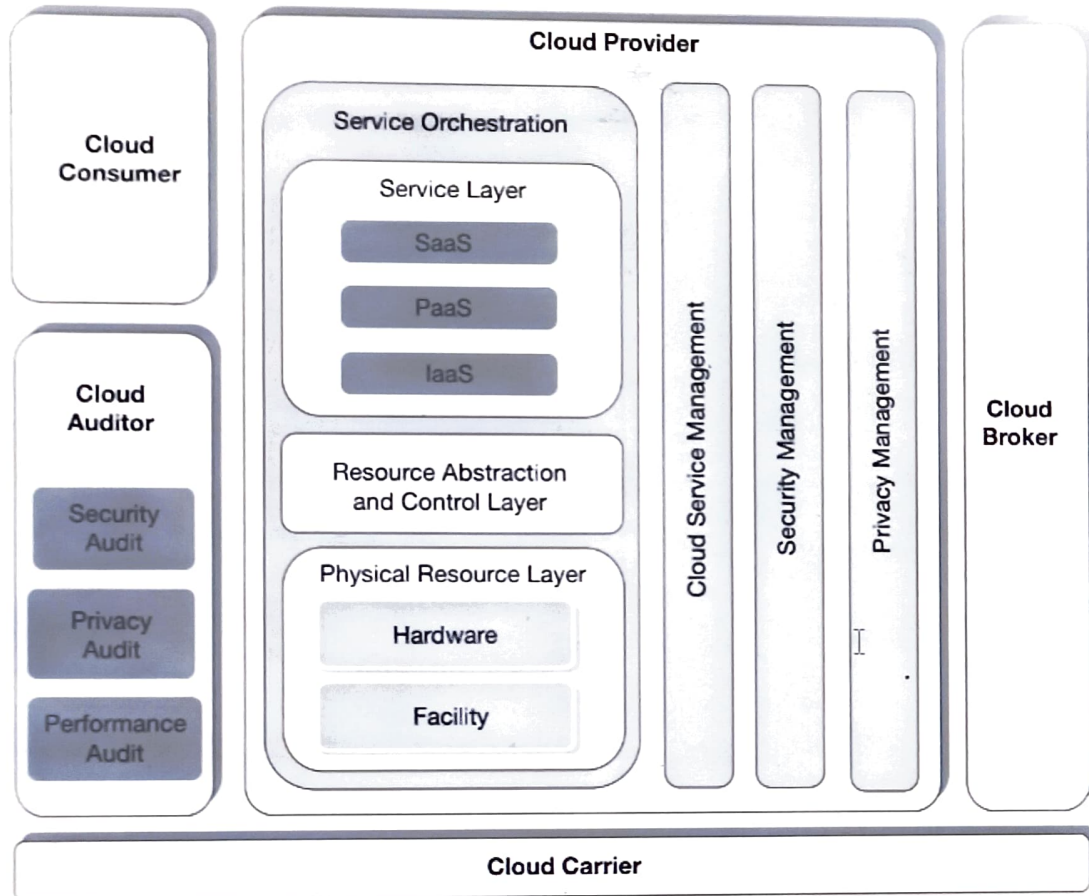


FIG 4.2: NIST cloud computing reference architecture²

Private cloud does not support multi-tenancy and thus cannot offer much in terms of efficiency. But, it ensures more security and privacy of data.

Table 4.2 A comparison between private cloud and public cloud

<i>Private Cloud</i>	<i>Public Cloud</i>
It can be both of types of on-premises and off-premises.	There cannot be any on-premises public cloud deployment.
On-premises private cloud can be delivered over the private network.	It can only be delivered over public network.
It does not support multi-tenancy feature for unrelated and external tenants.	It demonstrates multi-tenancy capability with its full ability.
The resources are for exclusive use of one consumer (generally an organization).	The resources are shared among multiple consumers.
A private cloud facility is accessible to a restricted number of people.	This facility is accessible to anyone.
This is for organizational use.	It can be used both by organization and the user.

Private Cloud

The consumers have important roles to play in management of the cloud.

The consumers have more control over the environment.

It provides more confidence regarding security of data as remains under the control of consumer organization's security boundary.

The cost of computing is more in comparison to public cloud.

It is not an ideal scenario for promoting pay-as-you-use philosophy.

The resource may often remain idle resulting in the resource wastage.

It is not so environment friendly like public cloud.

Public Cloud

The management is entirely provider's responsibility.

The consumers have very less control or no control.

The public cloud deployment often creates concerns regarding security and privacy of data.

This is more economical as multiple unrelated consumers (tenants) share same infrastructure.

It is an ideal model for practicing pay-as-you-use philosophy.

The resource utilization is optimum due to the presence of larger number of consumers.

It promotes green computing at its best.

4.2.3 Community Cloud

The *community cloud* deployment model allows access to a number of organizations or consumers belonging to a community and the model is built to serve some common and specific purpose. It is for the use of some community of people or organizations who share common

Table 4.3 A comparison between private cloud and community cloud

<i>Private Cloud</i>	<i>Community Cloud</i>
One private cloud can be used by one consumer (may be an organization).	One community cloud can be used by one community.
Access is restricted among members of a single consumer.	Access is restricted among members of a single community.
Resources of a private cloud are for exclusive use of one consumer.	Resources are shared among multiple consumers of a single community.
It may reside at consumer's premises.	It may reside at some consumer's premises within the community.
The private cloud (on-premises) can be delivered over private network.	To provide access to multiple consumers (of a community), it delivers over public network.
It does not support multi-tenancy.	Here multi-tenancy is supported.
This is for organizational use.	This one is for use of the communities.

4.2.4 Hybrid Cloud

A *hybrid cloud* is generally created by combining private or community deployment with public cloud deployment together. This deployment model helps businesses to take advantage of private or community cloud by storing critical applications and data. There at the same time, it provides the cost benefit by keeping shared data and applications on the public cloud. Figure 4.9 demonstrates a hybrid cloud model combining public cloud with on-premises

to take the responsibility of managing all of the underlying layers. For instance, a software-developing company needs to arrange and manage the required computing infrastructure

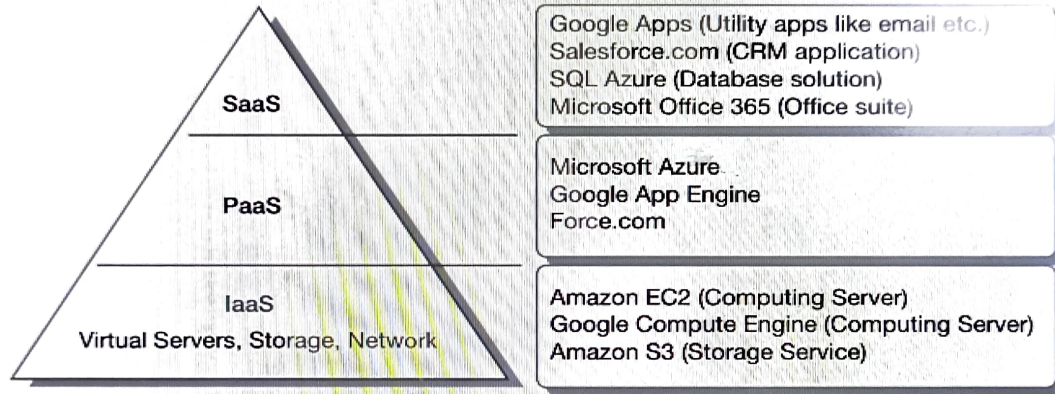


FIG 5.8: The layered cloud service model

... is delivered as a service (PaaS) and the service provider manages everything up to the platform. The whole system remains transparent to the users.

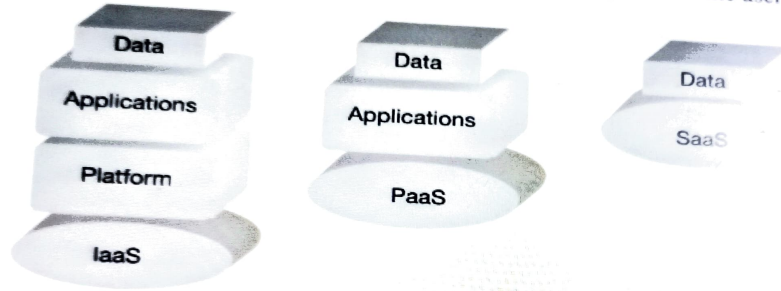


FIG 5.10: The working model of cloud computing system

	Application Service Provider (ASP)	Software-as-a-Service (SaaS)
Application development	ASPs are generally not the developer of the applications they host.	In SaaS model, the providers are the developer of the applications.
Deployment and management	ASPs deploy and manage the applications.	Service providers of the application themselves manage everything.
Multi-tenancy	One deployment can be used by a single consumer/organization only. Multi-tenancy is not supported.	Applications are designed to be used in multi-tenant environments.

(continued)

Table 5.1 (continued)

Scalability	Applications cannot grow or shrink with demand. They are not scalable.	It delivers scalable applications which can grow or shrink with changing demand.
Customization	ASPs are dependent on the commercial owner of application for any customization of the application.	Providers are the owner of the applications and hence customization is easier.
Domain knowledge	ASPs may not have much domain knowledge of application they host.	SaaS providers are specialists in their respective application domains.
Economy of cost	It incurs less cost than the traditional on-premises application model.	It is more economic because of the multi-tenancy and scaling ability.

Sourcing can either be outsourced or insourced for each of the eight cloud forms discussed above. In Jericho Forum's Cloud Cube Model, this *fourth dimension* is represented by two different colours for painting the cubes. Hence the eight smaller cubes that come out after combining the first three dimensions discussed above can take either of the two colours as shown in Figure 6.4

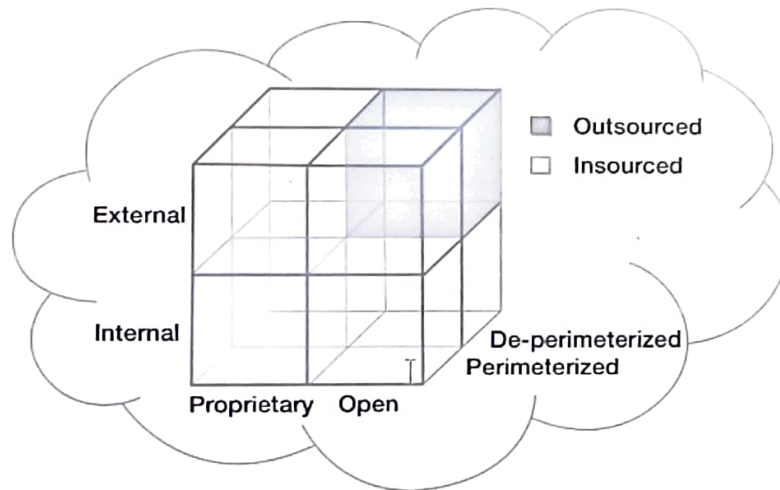


FIG 6.4: Jericho Forum's Cloud Cube Model⁷

It (figure 6.4) represents the cloud cube model after combining the four security dimensions together. In this model, the top-right-rear E / O / D-p cloud formation is considered as the

Virtualized physical server can host multiple virtual machines, where each one of them can have a different OS.

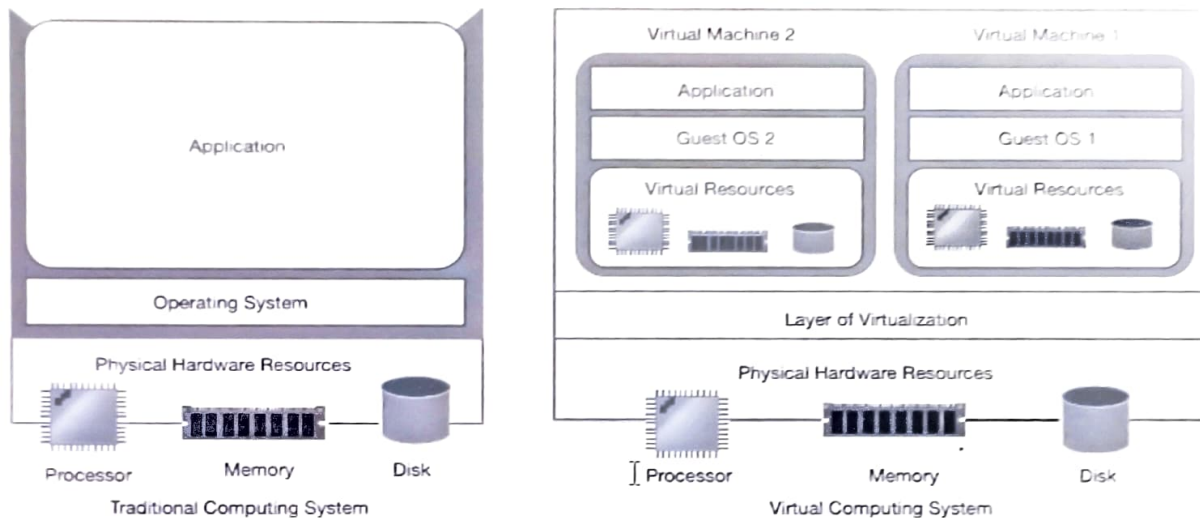


FIG 7.3: Conventional computing system versus Virtualized computing system

Table 7.1 Comparison between non-virtualized and virtualized machine environments

Non-Virtualized Machine Environment

At a moment, one single OS can run on a physical machine.

Application and hardware system remain tightly coupled.

Virtualized Machine Environment

Multiple OS can run simultaneously on one physical machine.

Virtual Machines isolates applications from the underlying hardware.



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Table 7.1 Comparison between non-virtualized and virtualized machine environments.

Non-Virtualized Machine Environment	Virtualized Machine Environment
At a moment, one single OS can run on a physical machine.	Multiple OS can run simultaneously on one physical machine.
Application and hardware system remain tightly coupled.	Virtual Machines isolates applications from the underlying hardware.

Resource utilization rate is low in most of the times.

These increase cost of business due to low resource utilization.

They have the inflexible approach.

Resource utilization improves as multiple VMs share same set of physical resources.

They are cost-effective if planned properly.

They provide lot of flexibility to system designers.

7.5.1 The Layer of Virtualization

From Figure 7.3, it can be seen that virtual machines are created over the virtualization layers. This virtualization layer is actually a set of control programs that creates the environment for the virtual machines to run on. This layer provides the access to the system resources to the virtual machines. It also controls and monitors the execution of the virtual machines over it. This software layer is referred as the *Hypervisor* or *Virtual Machine Monitor (VMM)*.



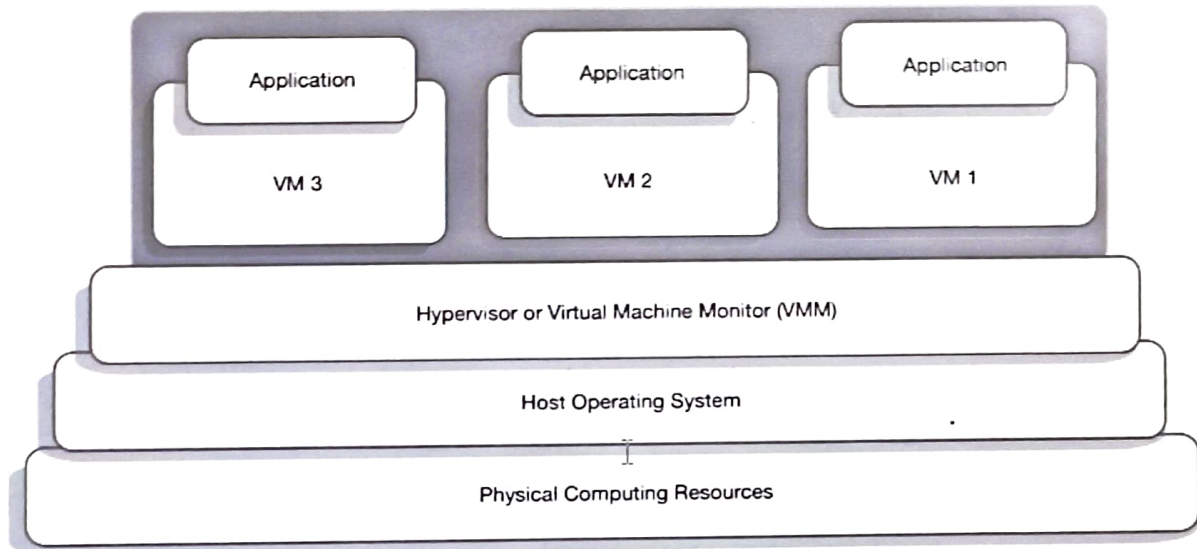


FIG 7.4: A model of hosted machine virtualization approach

Drawbacks: A hosted hypervisor does not have direct access to the hardware resources and hence, all of the requests from virtual machines must go through the host OS. This may degrade the performance of the virtual machines.

Another drawback of the hosted virtualization is the lack of support for real-time operating systems. Since the underlying host OS controls the scheduling of jobs it becomes unrealistic to run a real-time OS inside a VM using hosted virtualization.

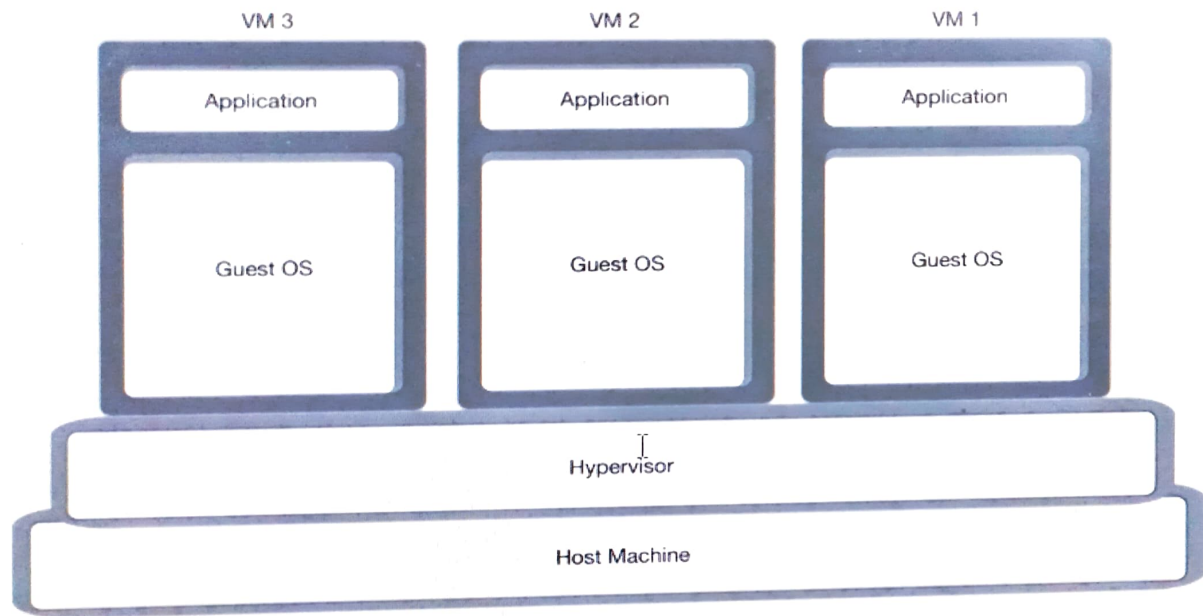


FIG 7.6: A model of full virtualization

remains completely isolated from physical resource layers by the hypervisor. This provides flexibility as almost all of the available operating systems can work as guest OS. VMware's virtualization product *VMWare ESXi Server* and *Microsoft Virtual Server* are few examples of full virtualization solution.

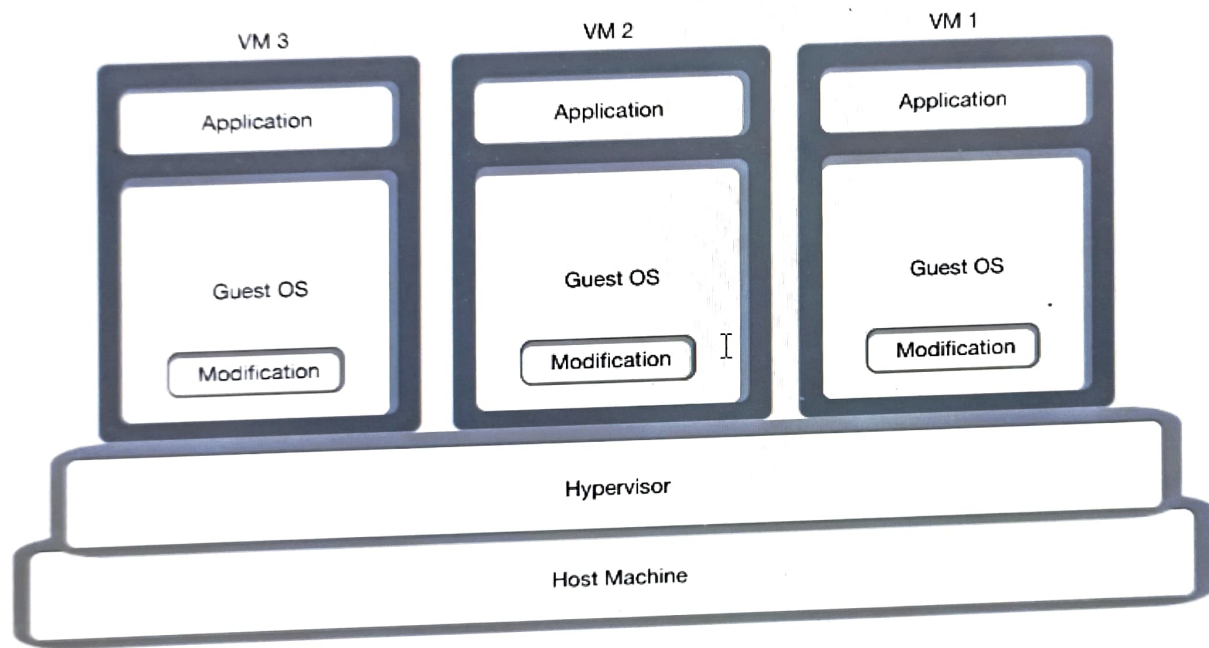


FIG 7.7: A model of para-virtualization

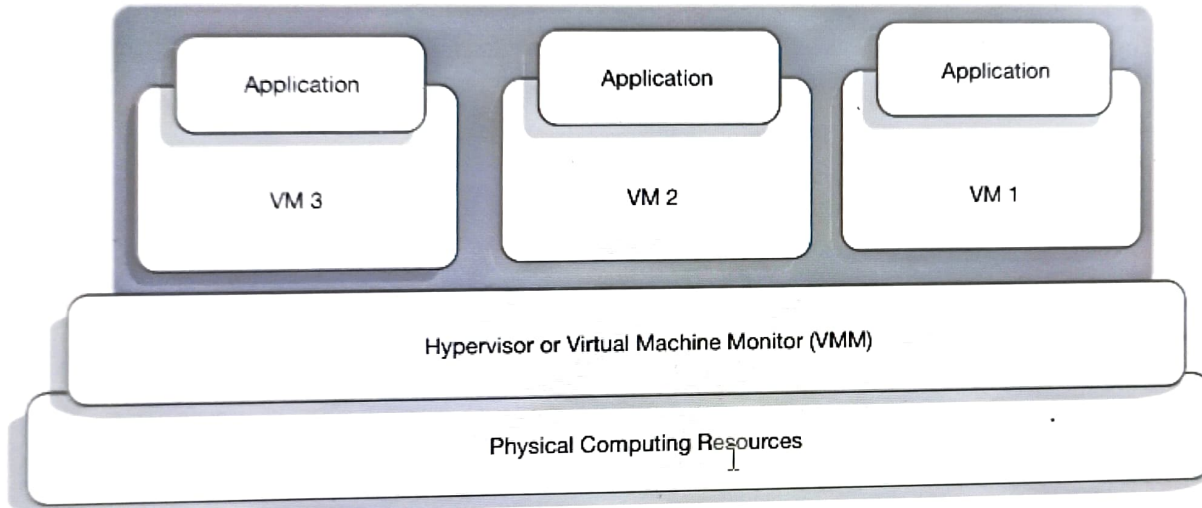


FIG 7.5: A model for the bare metal approach of machine virtualization

Type 1 or bare metal hypervisor does not use any host operating system.

7.6 EXPLORING HYPERVISOR OR VIRTUAL MACHINE MONITOR

Table 7.3 Comparison among Full Virtualization, Para-Virtualization and Hardware-Assisted Virtualization

Full Virtualization	Para-Virtualization or OS-Assisted Virtualization	Hardware-Assisted Virtualization
Guest OS has no role in virtualization.	Guest OS plays role in virtualization.	Guest OS has no role in virtualization.
Guest OS remains unaware about the virtualization.	Guest OS has to be aware about the virtualization.	Guest OS remains unaware about the virtualization.
Normal version of available OS can be used as guest OS.	Modified version of available OS is required.	Normal version of available OS can be used as guest OS.
It provides good options for guest OS.	It provides lesser options for guest OS.	It provides good options for guest OS.
Guest OS is not hypervisor-specific.	Guest OS is tailored to be hypervisor-specific.	Guest OS is not hypervisor-specific.
Here it requires no special feature in the host CPU.	Here it requires no special feature in the host CPU.	Here it requires explicit features in the host CPU.
Hardware does not play role in virtualization.	Hardware does not play role in virtualization.	Hardware plays role in virtualization.
Hypervisor takes care of all of the virtualization tasks.	Guest OS along with hypervisor take care of the virtualization tasks.	Specialized hardware device along with hypervisor take care of virtualization tasks.
Virtualization overhead of hypervisor is more.	Virtualization overhead of hypervisor is less.	Virtualization overhead of hypervisor is less.
Virtualization performance is little slow.	Virtualization performance is better.	Virtualization performance is better.
It provides high level of security as all of the virtualization controls remain with the hypervisor.	Here the security is compromised as guest OS has some control in virtualization.	Here the security is compromised as calls from guest OS can directly access the hardware.